

Question Set

Q1. A monthly income of a group of 10,000 persons were found to be normally distributed with mean 750 rs and SD rupee 50. Show that of this group about 95% had income exceeding rupee 668 and only 5% has income exceeding rupees 832. What is the lowest income among the richest 100?

Q2. Suppose X has $N(3,4)$. Find c , s.t $P(X > c) = 2 P(X \leq c)$.

Q3. The time it takes for a courier to deliver a package follows normal distribution with mean $\mu = 30$ minutes and $\sigma = 5$ minutes. The company guarantees that if a delivery takes longer than T minutes, they will refund the fee. If they want to refund only the slowest 15% of deliveries, what should T be?

Q4. If $x \sim N(75, 25)$. Find $P(x > 80 | x > 77)$.

Q5. Diameter of an electric cable is normally distributed with mean 0.8 and variance 0.0004. What is probability that diameter exceeds 0.81 inches?